

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

July 20, 2026

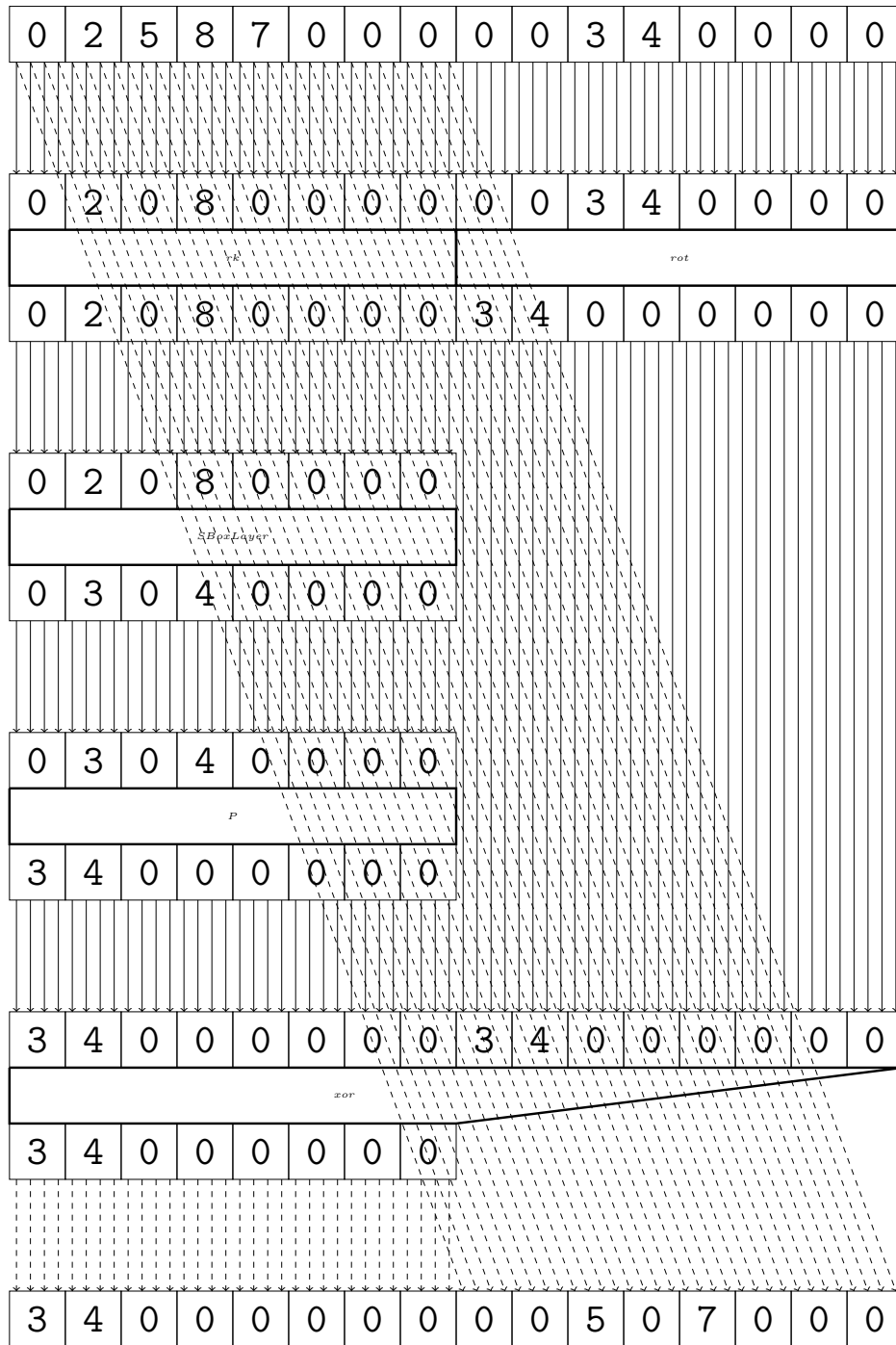
1 LBlock

Maximal linear correlation: 2^{-18}

0	2	5	8	7	0	0	0	0	0	3	4	0	0	0	0
0	2	5	8	7	0	0	0	0	0	3	4	0	0	0	0
LBlock round															
3	4	0	0	0	0	0	0	0	0	5	0	7	0	0	0
3	4	0	0	0	0	0	0	0	0	5	0	7	0	0	0
LBlock round															
5	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0
LBlock round															
0	0	0	0	0	0	0	0	5	0	7	0	0	0	0	0
0	0	0	0	0	0	0	0	5	0	7	0	0	0	0	0
LBlock round															
7	0	0	0	0	0	5	0	0	b	0	0	2	0	0	0
7	0	0	0	0	0	5	0	0	b	0	0	2	0	0	0
LBlock round															
0	0	2	0	0	0	0	b	0	0	0	0	0	0	0	0
0	0	2	0	0	0	0	b	0	0	0	0	0	0	0	0
LBlock round															
0	0	0	0	0	0	0	0	0	2	0	0	0	0	b	
0	0	0	0	0	0	0	0	0	2	0	0	0	0	b	
LBlock round															
2	0	0	0	0	b	0	0	0	f	0	0	0	0	0	8
2	0	0	0	0	b	0	0	0	f	0	0	0	0	0	8
LBlock round															
0	0	0	0	0	8	0	f	2	0	0	0	0	b	c	1
0	0	0	0	0	8	0	f	2	0	0	0	0	b	c	1
LBlock round															
0	0	0	b	c	1	2	0	0	0	c	0	8	0	0	0
0	0	0	b	c	1	2	0	0	0	c	0	8	0	0	0
LBlock round															
c	0	8	0	0	0	0	0	8	e	0	b	c	1	2	0
c	0	8	0	0	0	0	0	8	e	0	b	c	1	2	0
Final round															
8	e	0	b	c	1	2	0	c	0	8	0	0	0	0	0
8	e	0	b	c	1	2	0	c	0	8	0	0	0	0	0

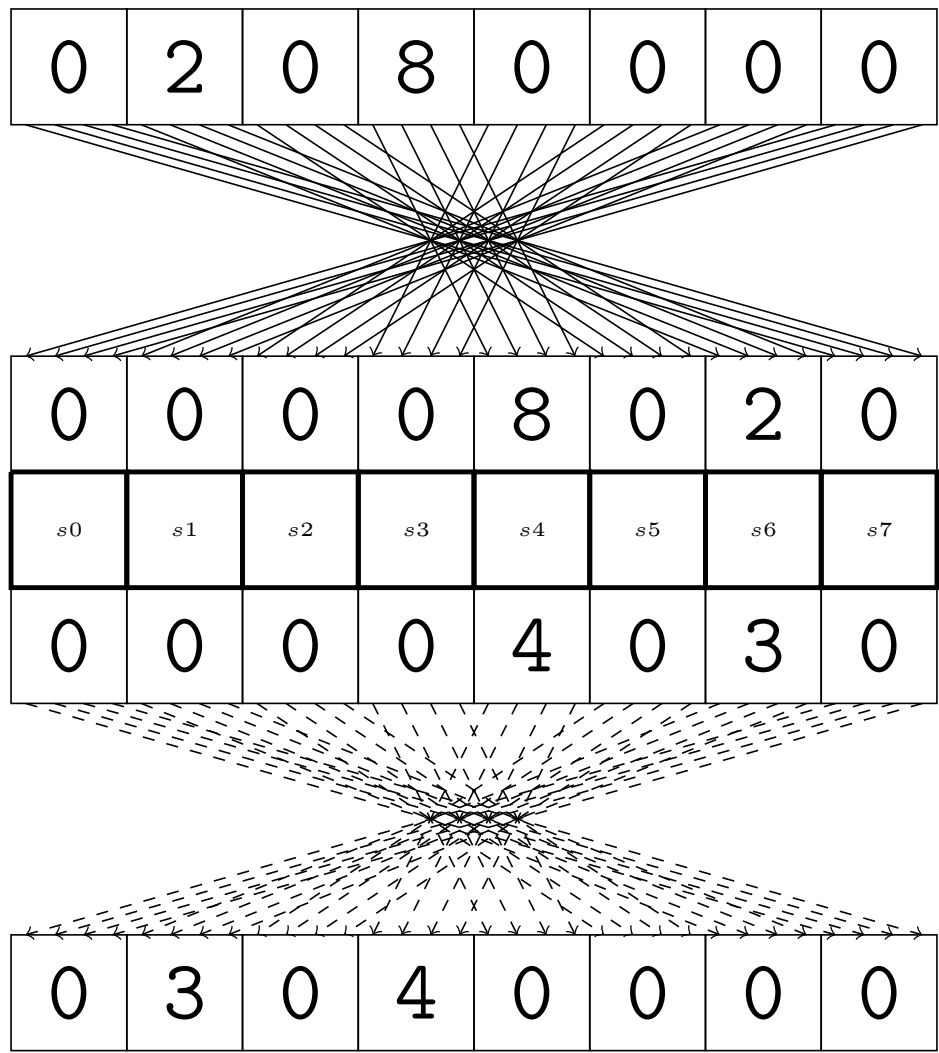
2 LBlock_round

Maximal linear correlation: $2^{-2.0}$



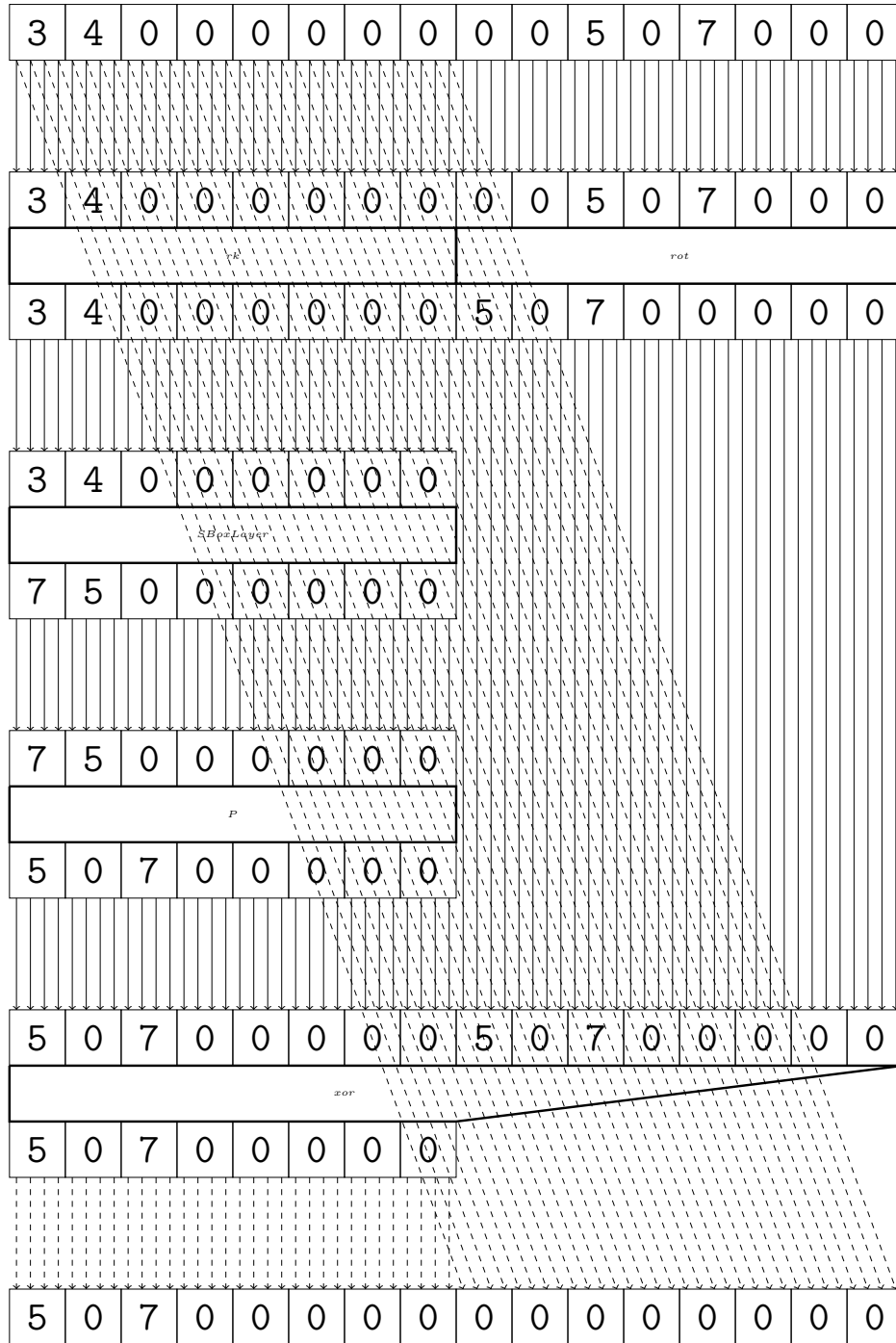
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



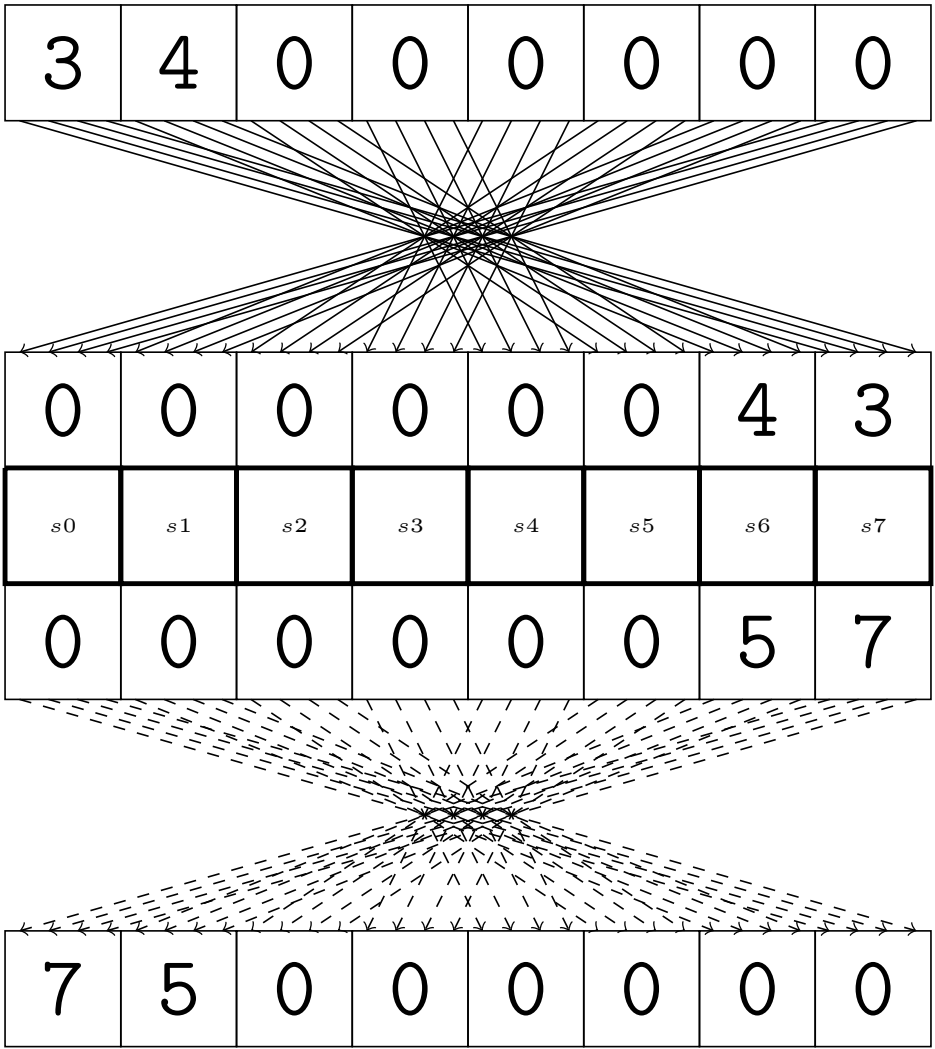
4 LBlock_round

Maximal linear correlation: $2^{-2.0}$



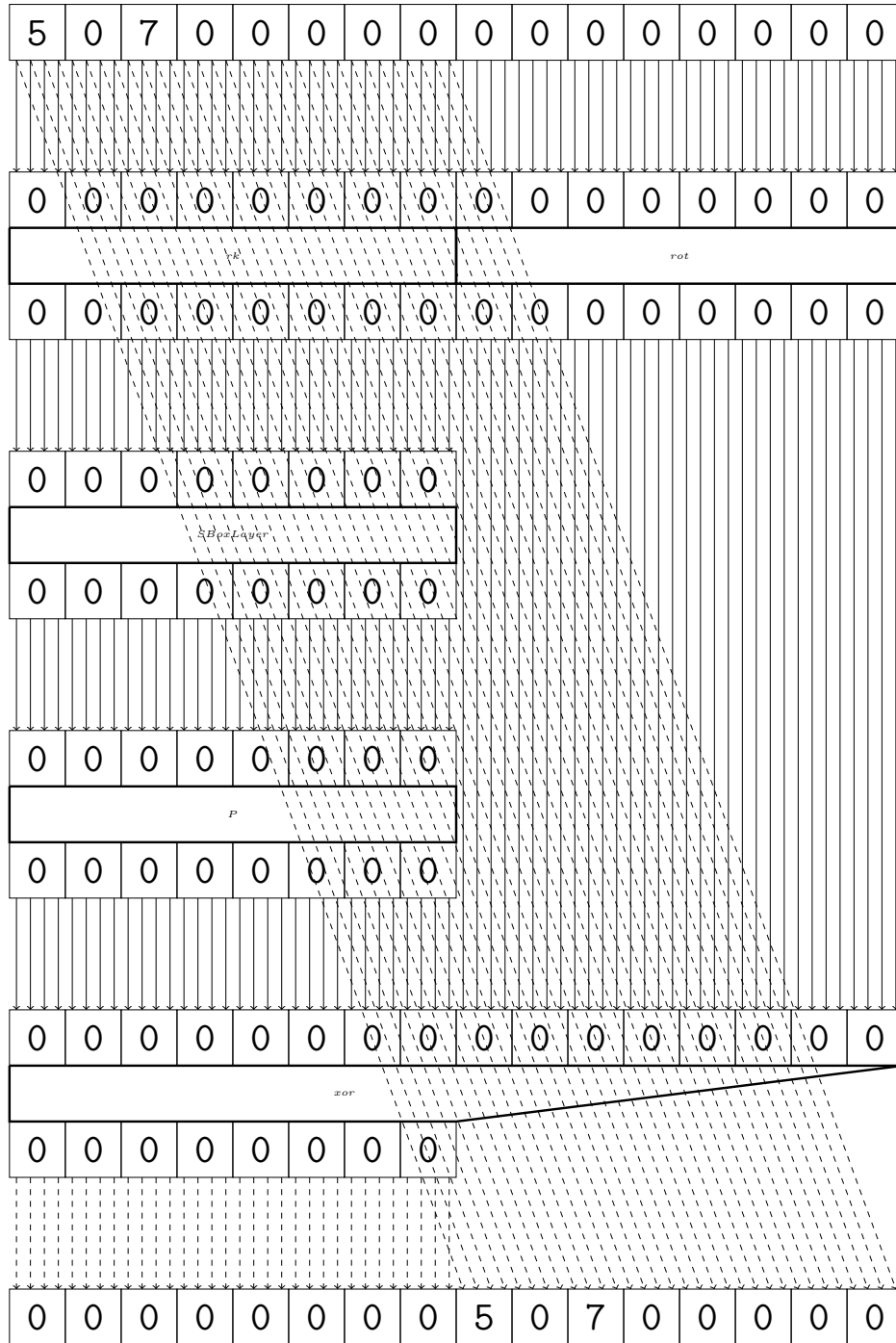
5 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



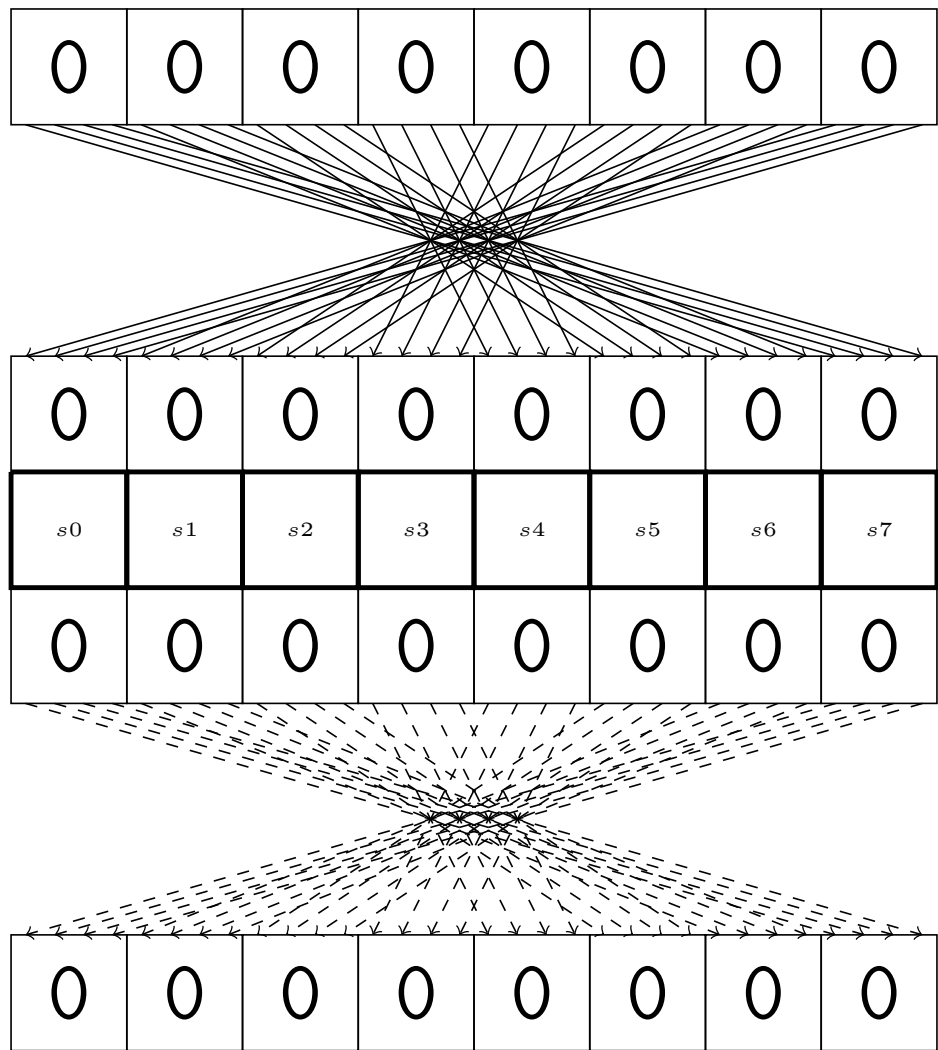
6 LBlock_round

Maximal linear correlation: $2^{-0.0}$



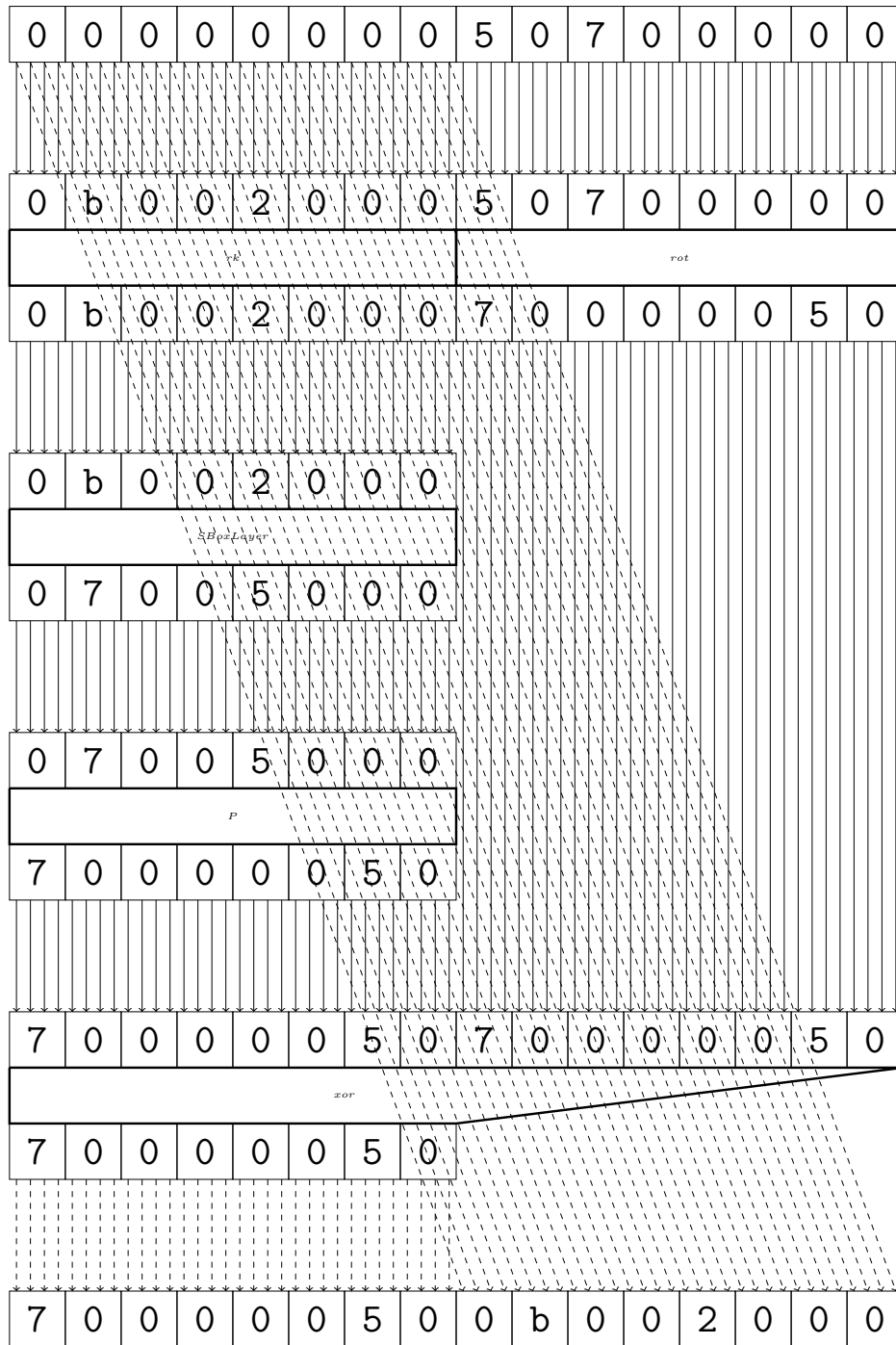
7 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



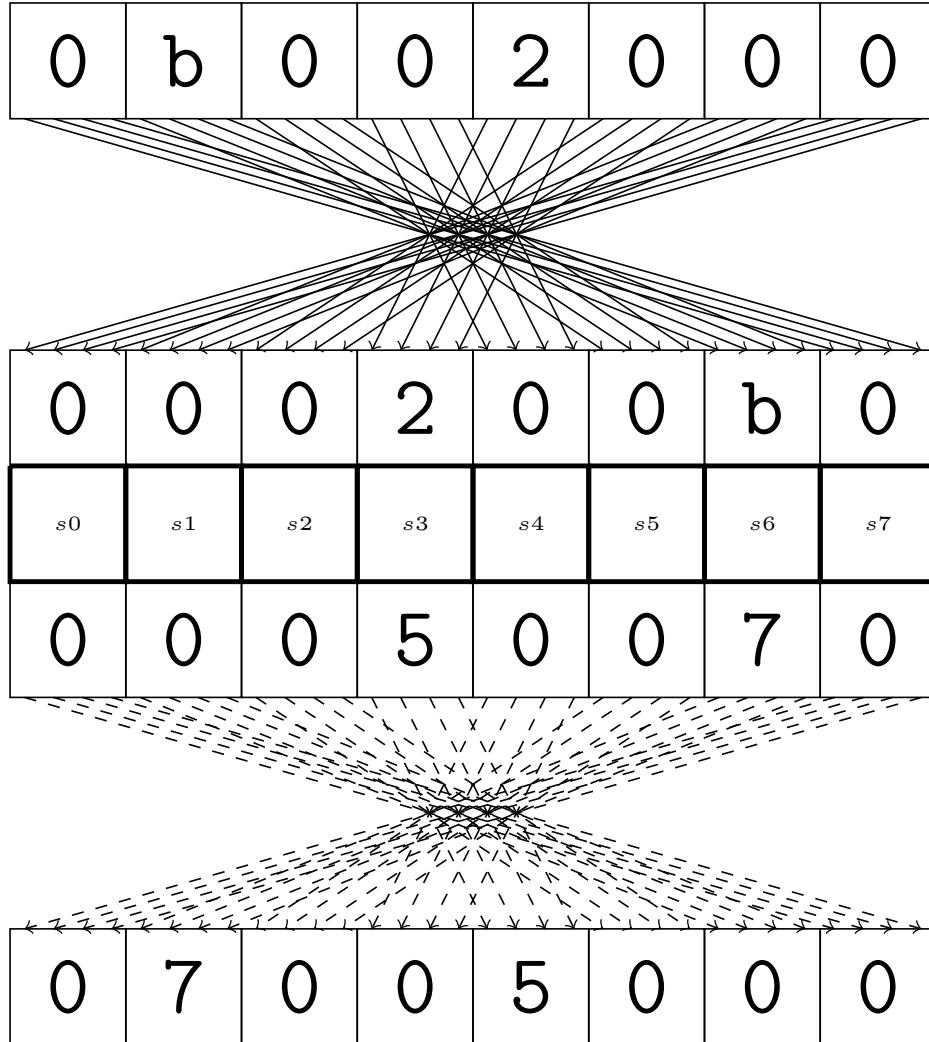
8 LBlock_round

Maximal linear correlation: $2^{-2.0}$



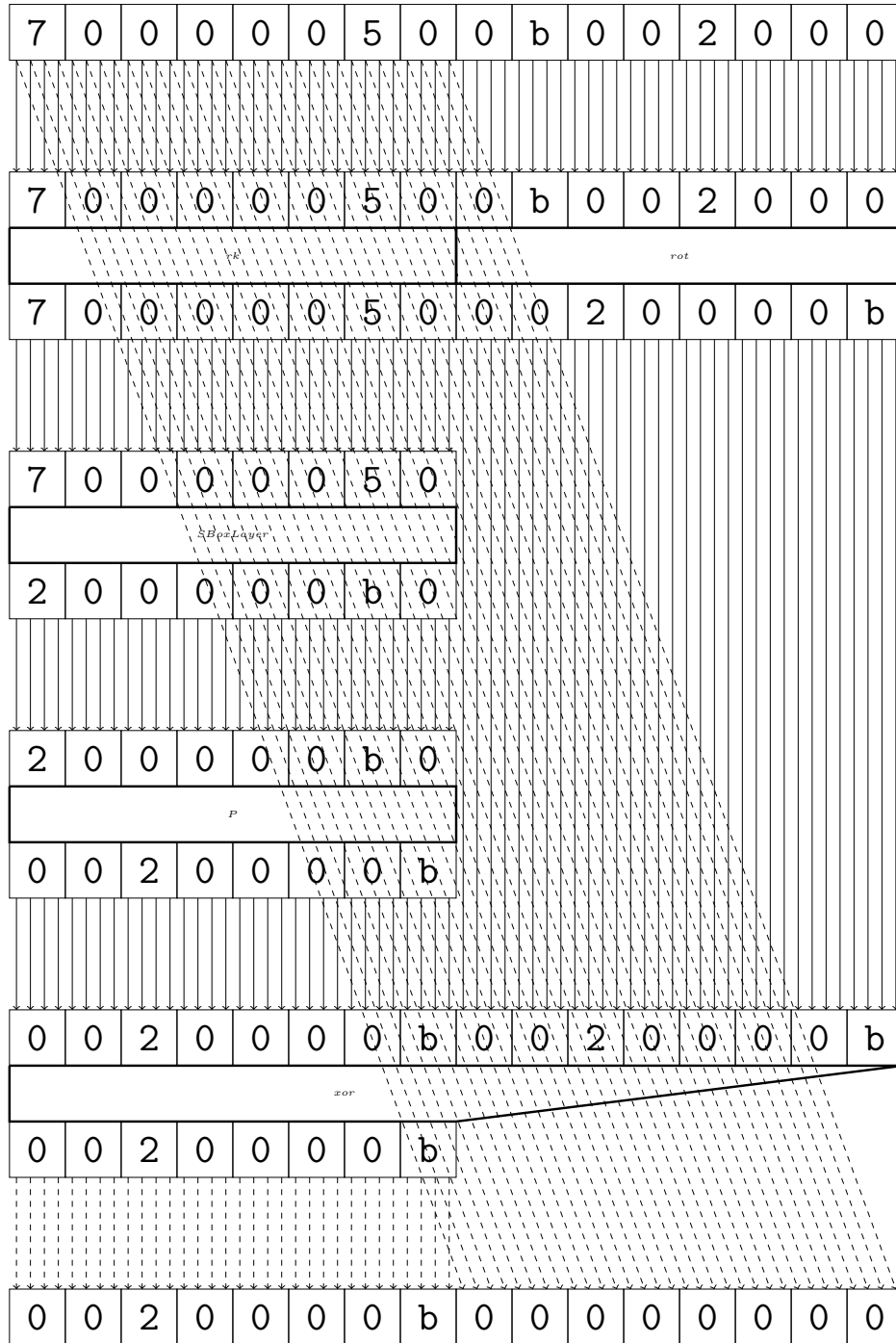
9 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



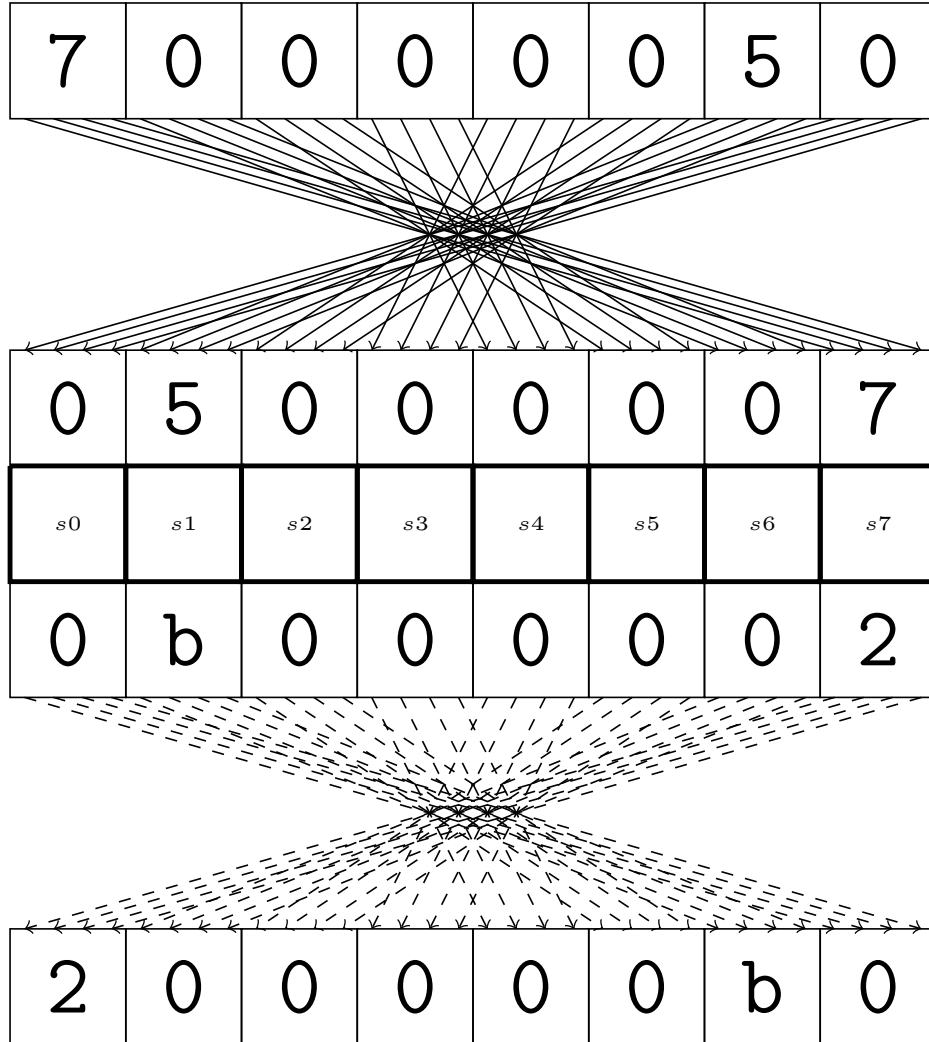
10 LBlock_round

Maximal linear correlation: $2^{-2.0}$

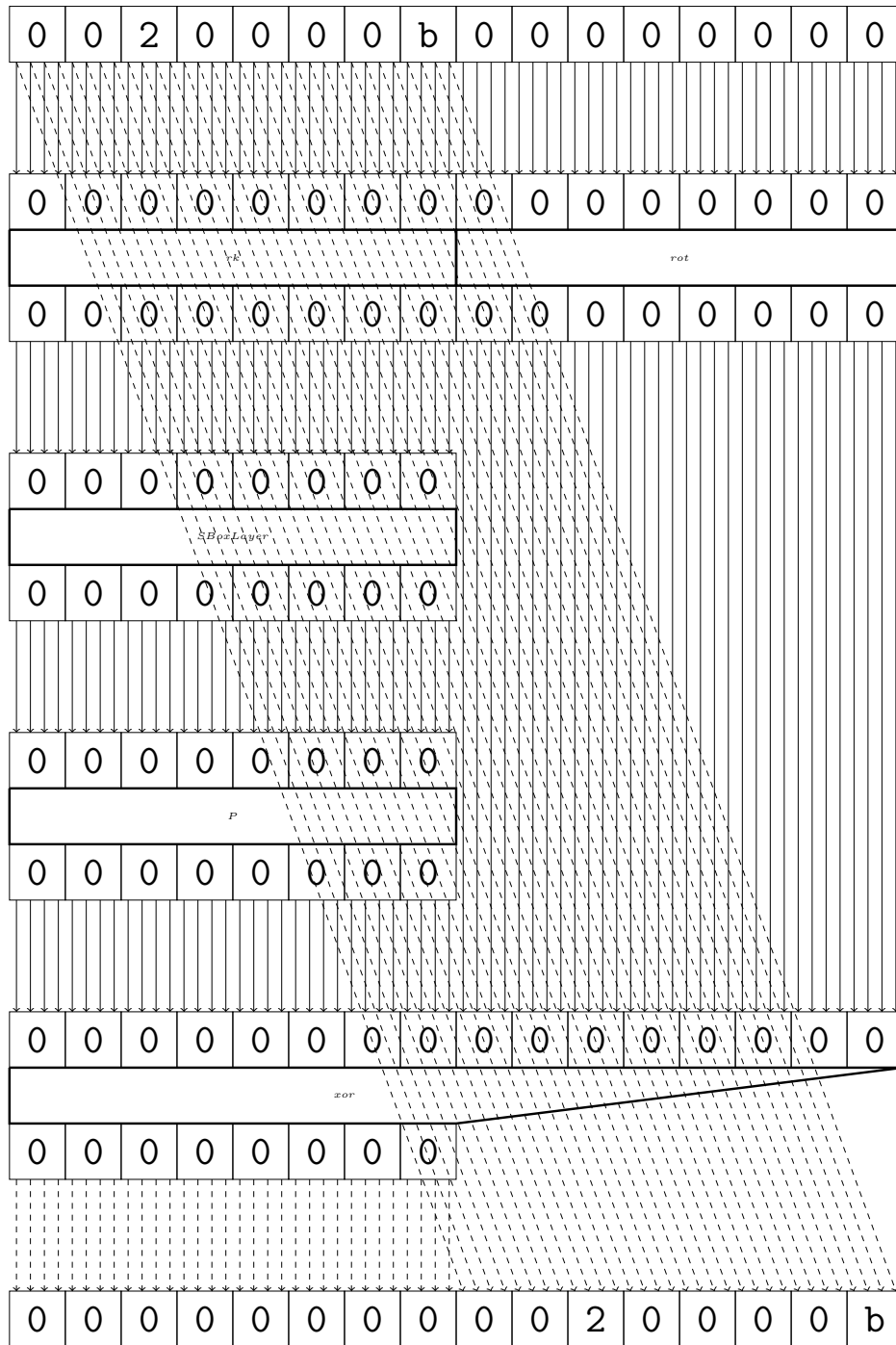


11 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

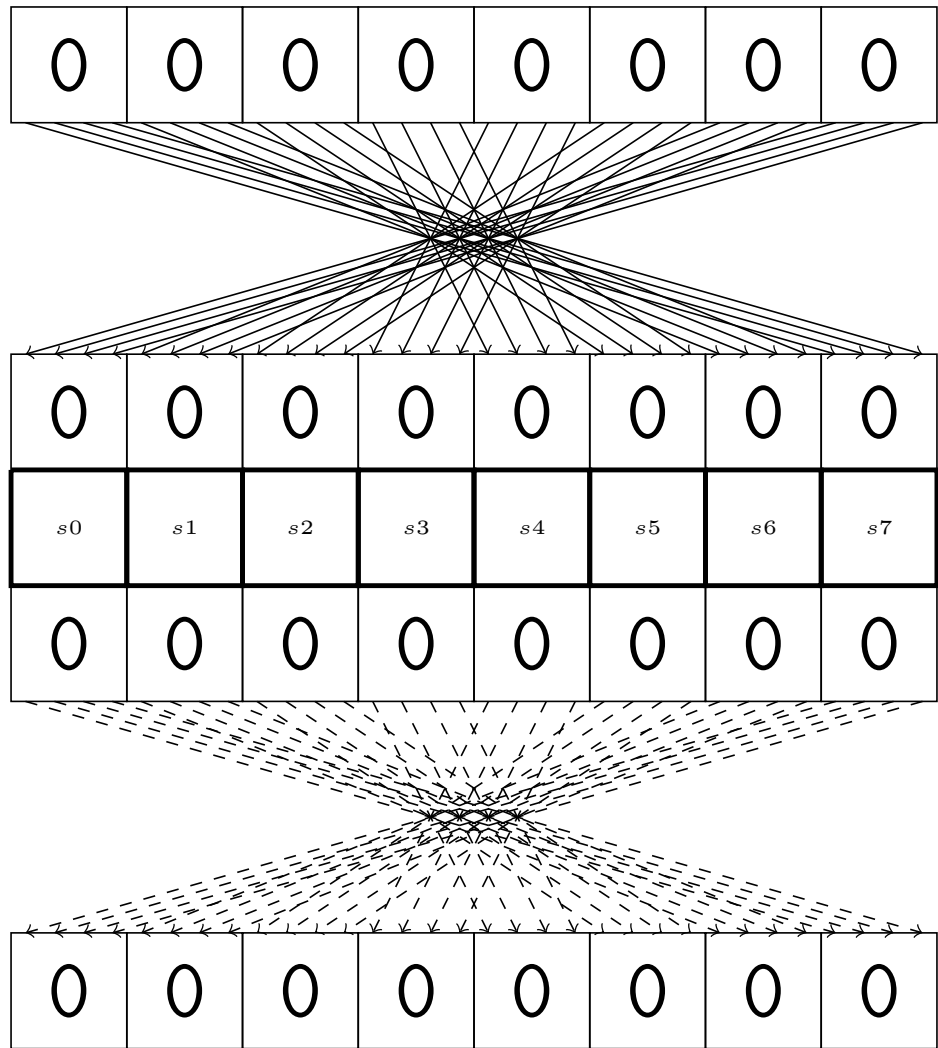


12 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

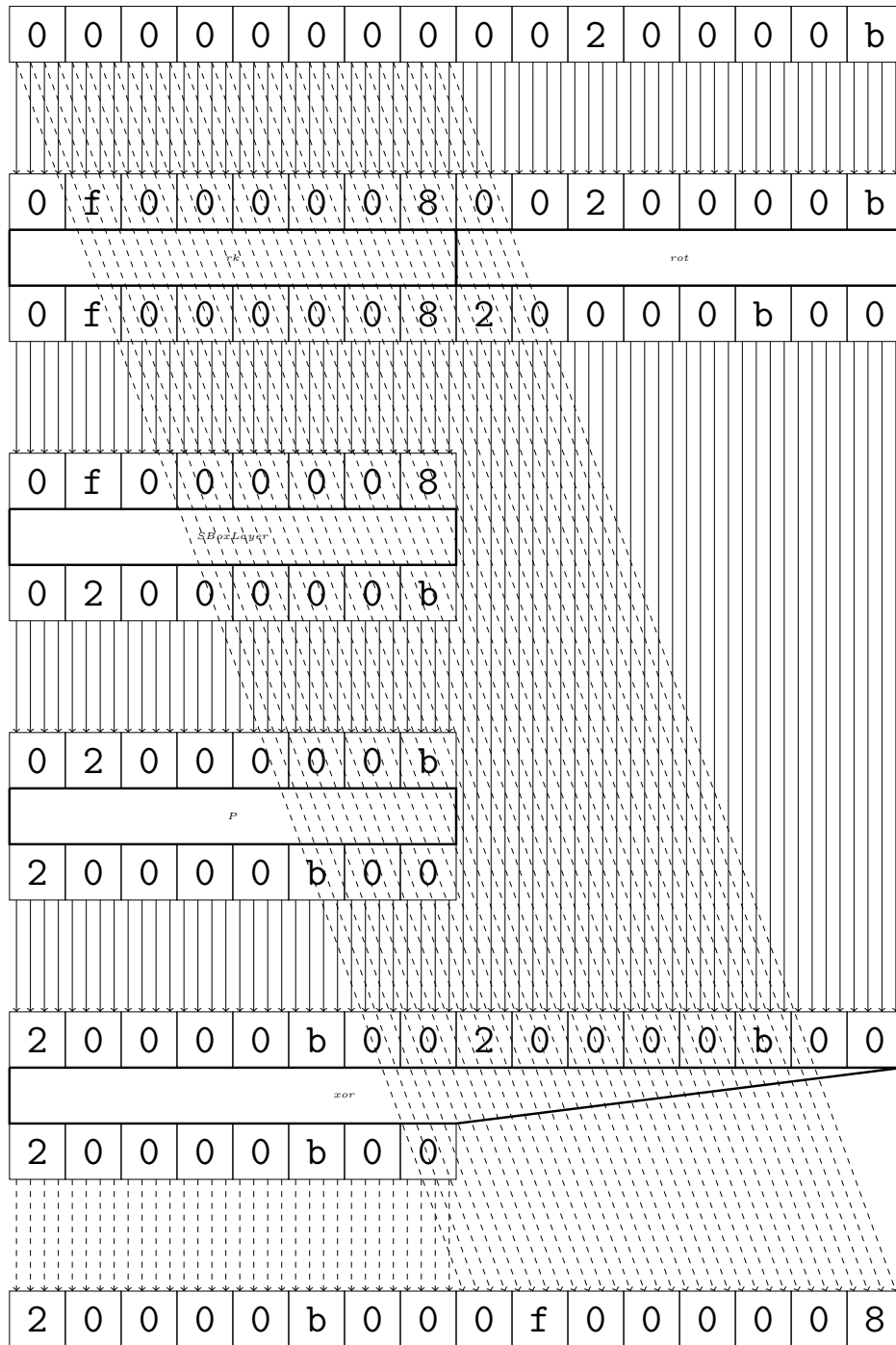
13 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



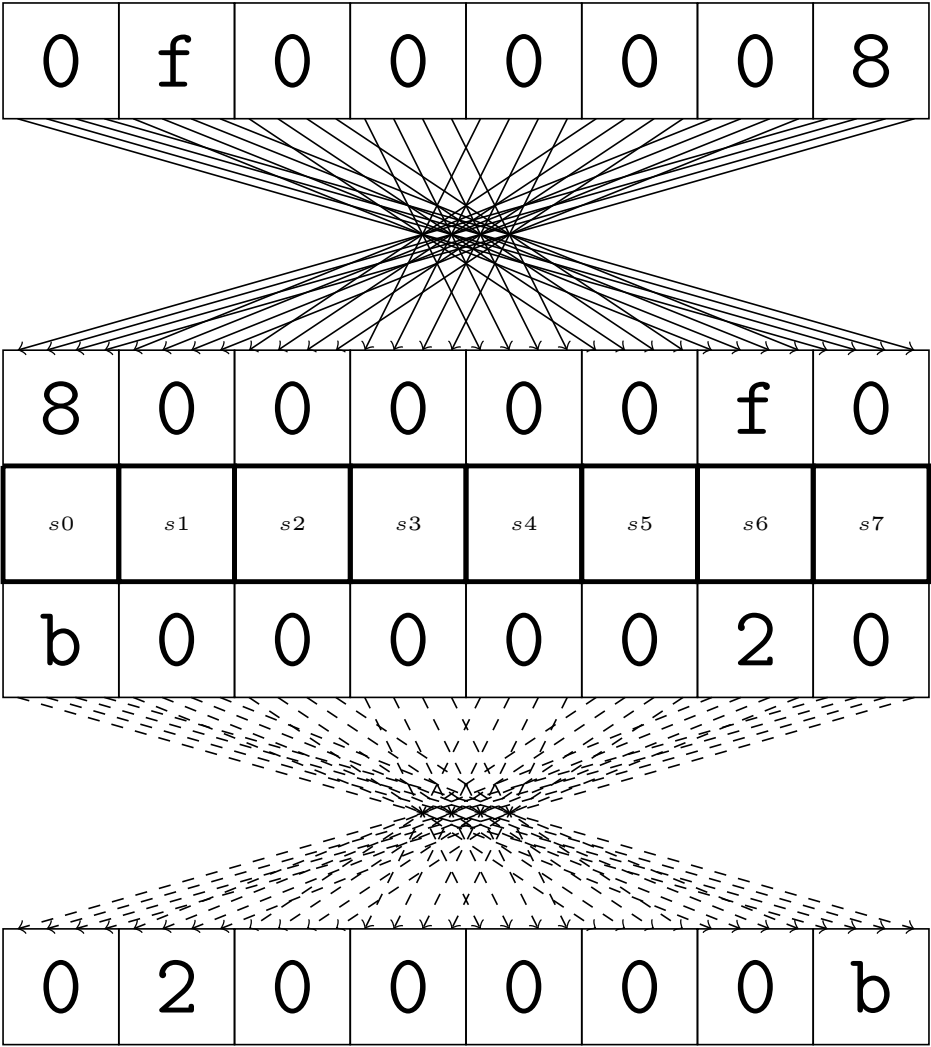
14 LBlock_round

Maximal linear correlation: $2^{-2.0}$

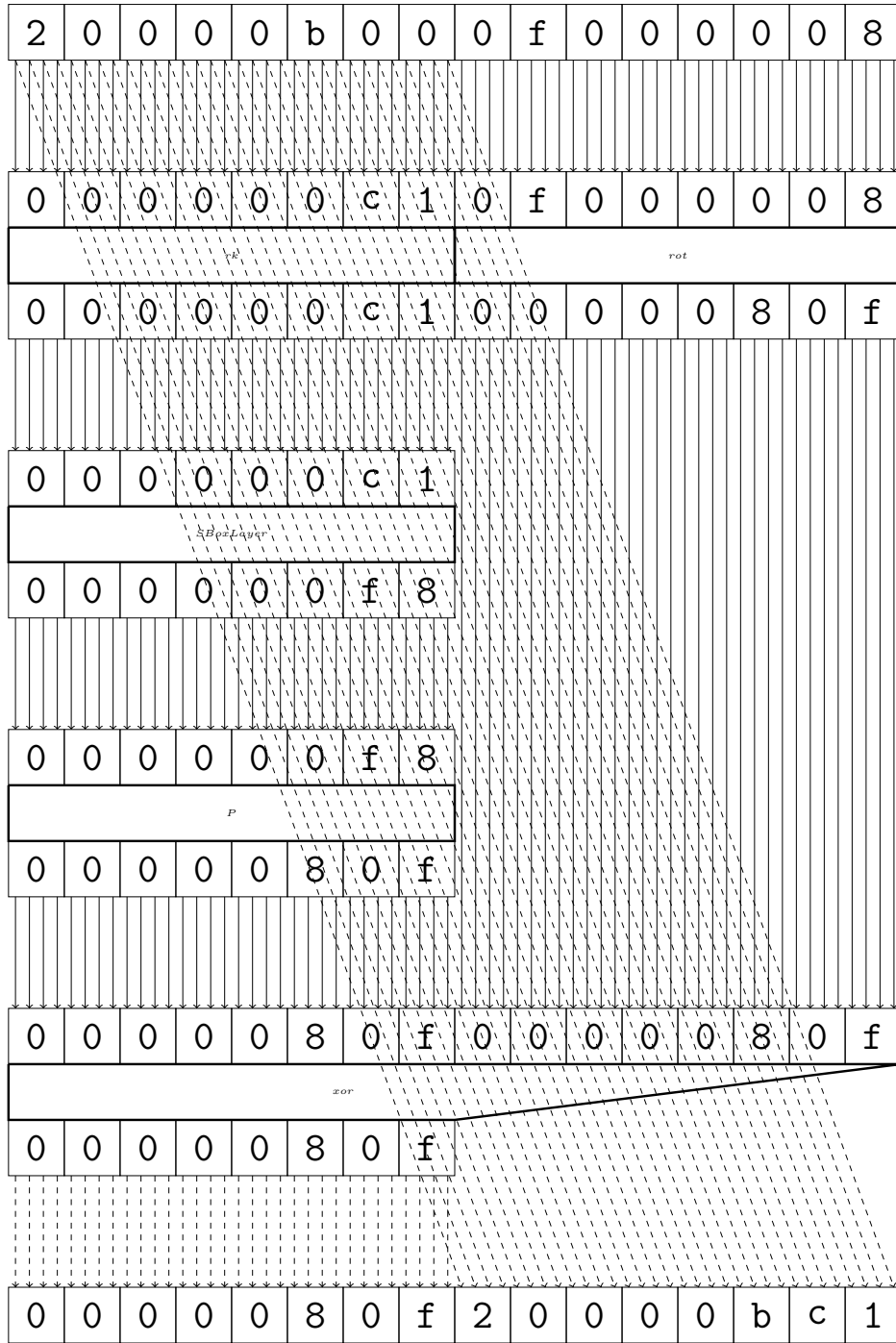


15 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

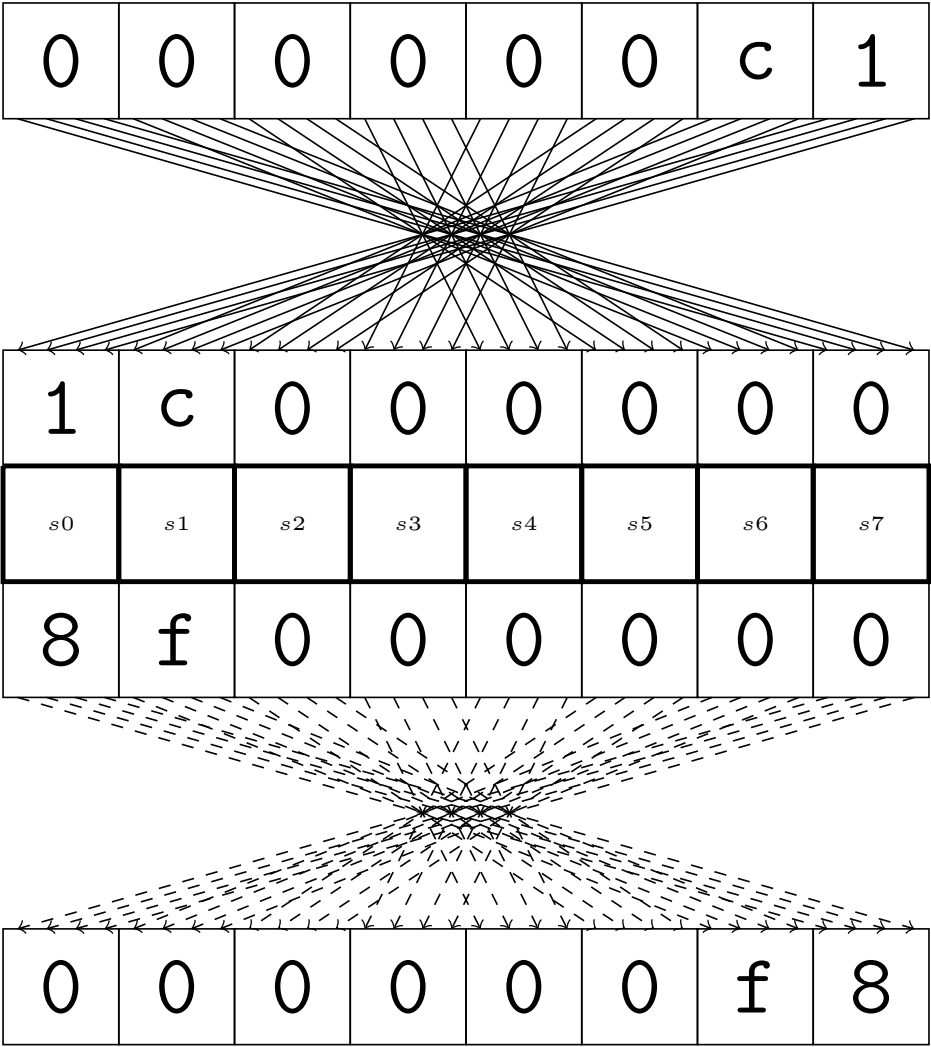


16 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

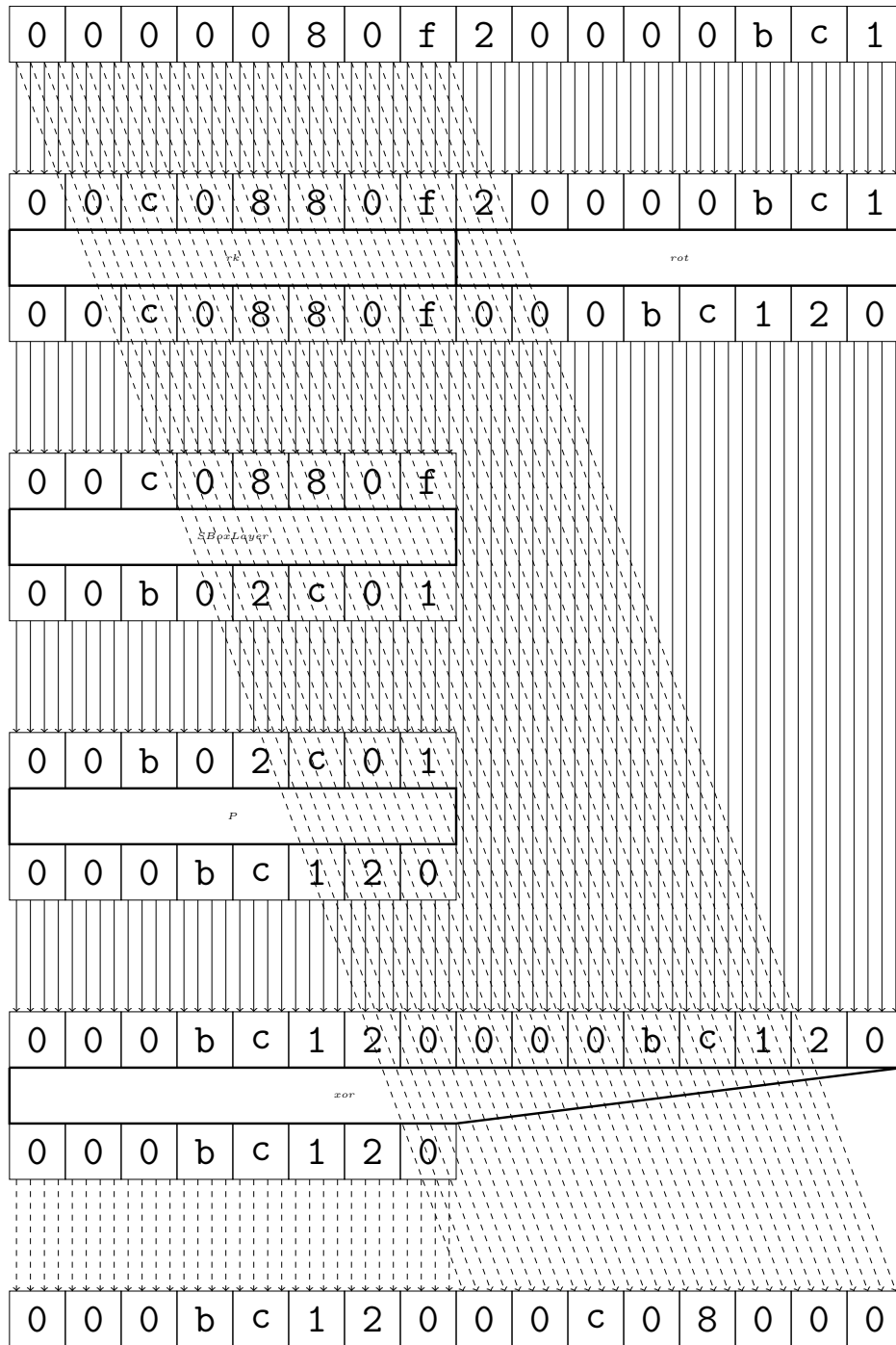
17 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



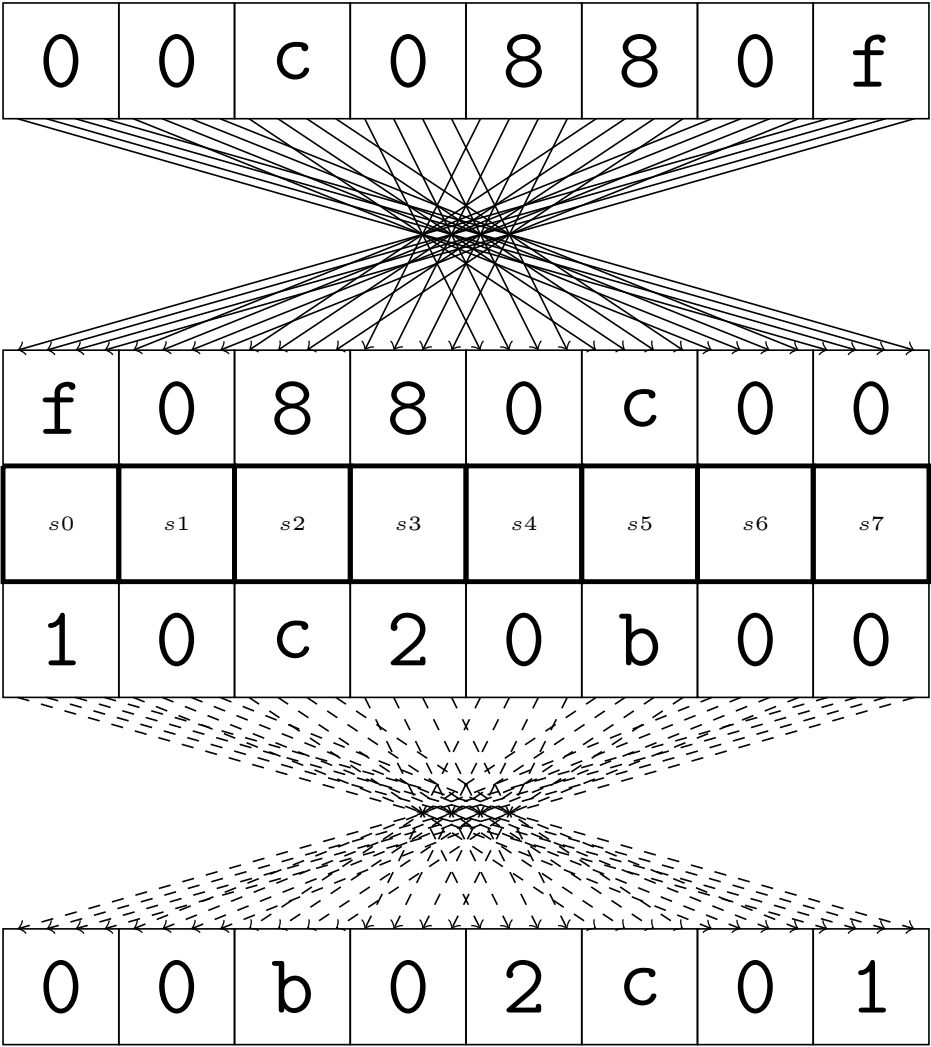
18 LBlock_round

Maximal linear correlation: $2^{-4.0}$



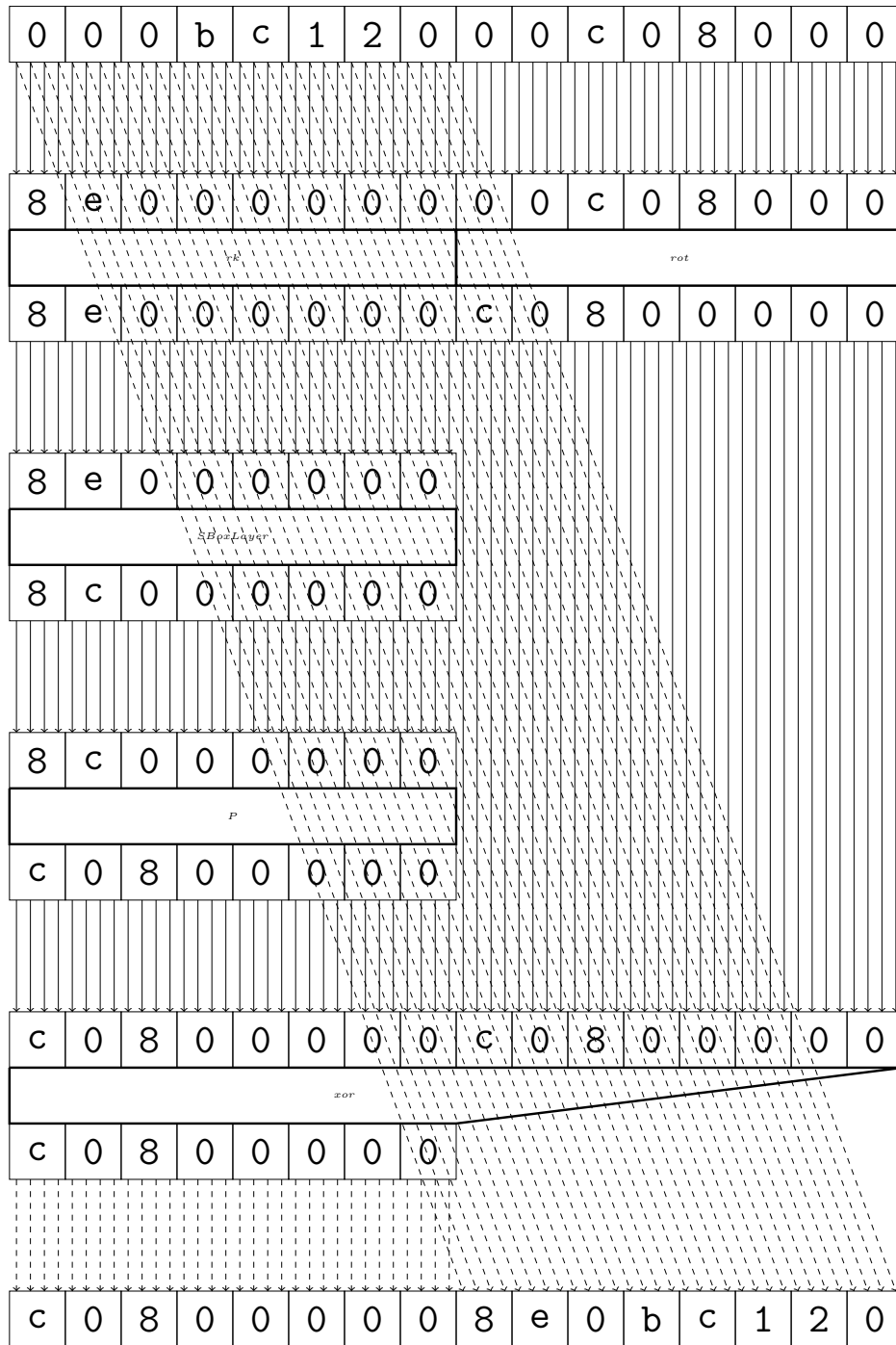
19 SBoxLayer

Maximal linear correlation: $2^{-4.0}$



20 LBlock_round

Maximal linear correlation: $2^{-2.0}$



21 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

